

# ABDUL SAMAD

Software Engineer – AI Researcher

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## PROFESSIONAL SUMMARY

Software Engineer with 5 years of experience and a core focus on backend engineering, scalable architectures, and cloud infrastructure. Currently engineering the high-throughput backend and DevOps behind Karachi's city-wide AI traffic-enforcement platform (TRACS). Also a published AI researcher, with peer-reviewed work on deep learning for medical imaging achieving 98% diagnostic accuracy. Skilled across Java/Spring Boot and the Node.js ecosystem, with a proven record of delivering reliable, high-performance systems.

## TECHNICAL SKILLS

**Languages:** Java, JavaScript, TypeScript, Python, SQL, HTML5, CSS3  
**Backend:** Spring Boot, Node.js, Express.js, Nest.js, Flask, FastAPI, Hibernate, RESTful APIs, GraphQL, Microservices  
**Frontend:** React.js, Next.js, Material UI, Android Native, React Native, React Native Expo, Flutter  
**Database:** MySQL, PostgreSQL, MongoDB, Oracle, Redis, SQLite  
**Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, Jenkins, GitLab CI/CD, GitHub Actions, Nginx, PM2, Vercel  
**Blockchain:** Hyperledger Fabric, Smart Contracts, Tokenization, Immutable Systems  
**AI/ML:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Streamlit, Grad-CAM, ResNet, CNNs, RAG, LangChain  
**Tools:** Git, GitHub, GitLab, RabbitMQ, Kafka, SonarCloud, OWASP ZAP, JUnit, Mocha, Chai, Postman

## PROFESSIONAL EXPERIENCE

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| <b>Software Engineer</b><br><i>GCS (Pvt) Ltd</i>                                                                                                                                                                                                                                                                                                                                                                                                         | Feb 2026 – Present<br><i>Karachi, Pakistan</i>  |
| <ul style="list-style-type: none"><li>Engineer on TRACS, Karachi's city-wide AI traffic-enforcement platform; lead backend and DevOps reliability work (full details under Projects).</li></ul>                                                                                                                                                                                                                                                          |                                                 |
| <b>Backend Engineer (Contract)</b><br><i>Intelik – Immutable Intelligence</i>                                                                                                                                                                                                                                                                                                                                                                            | Jun 2025 – Jan 2026<br><i>Karachi, Pakistan</i> |
| <ul style="list-style-type: none"><li>Architected Node.js, Express.js, and Nest.js backends with blockchain integration and CI/CD pipelines.</li><li>Delivered enterprise products including ILOS for UBL and the Blocks tokenization platform; built corporate sites intelik.net and mtechsolutions.pk.</li></ul>                                                                                                                                       |                                                 |
| <b>Backend Developer</b><br><i>Internee.pk</i>                                                                                                                                                                                                                                                                                                                                                                                                           | Apr 2025 – May 2025<br><i>Remote</i>            |
| <ul style="list-style-type: none"><li>Built backend infrastructure for an employee-management system using Node.js with RESTful APIs.</li></ul>                                                                                                                                                                                                                                                                                                          |                                                 |
| <b>Backend Developer</b><br><i>Vertex Vision</i>                                                                                                                                                                                                                                                                                                                                                                                                         | Mar 2024 – Mar 2025<br><i>Karachi, Pakistan</i> |
| <ul style="list-style-type: none"><li>Designed and developed scalable backend services and RESTful APIs for enterprise applications using Node.js and Express.js.</li><li>Built and optimized relational and NoSQL data models, improving query performance and overall system reliability.</li><li>Implemented authentication, authorization, and third-party API integrations; collaborated with frontend teams to ship production features.</li></ul> |                                                 |
| <b>Software Development Instructor</b><br><i>Centre of Success</i>                                                                                                                                                                                                                                                                                                                                                                                       | Sep 2022 – Nov 2024<br><i>Karachi, Pakistan</i> |
| <ul style="list-style-type: none"><li>Taught web, mobile, and desktop development and designed a hands-on curriculum on modern frameworks and best practices.</li></ul>                                                                                                                                                                                                                                                                                  |                                                 |
| <b>Freelance Software Developer</b><br><i>Self-Employed</i>                                                                                                                                                                                                                                                                                                                                                                                              | Mar 2021 – Present<br><i>Remote</i>             |
| <ul style="list-style-type: none"><li>Delivered 15+ custom solutions (CRMs, web and mobile apps); integrated 20+ third-party services with 99.9% uptime.</li></ul>                                                                                                                                                                                                                                                                                       |                                                 |
| <b>Database Manager</b><br><i>M&amp;P, Karachi Headquarters</i>                                                                                                                                                                                                                                                                                                                                                                                          | Jun 2022 – Sep 2022<br><i>Karachi, Pakistan</i> |
| <ul style="list-style-type: none"><li>Managed database systems, ensuring data integrity, security, and performance with backup and recovery operations.</li></ul>                                                                                                                                                                                                                                                                                        |                                                 |

KEY PROJECTS

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| <b>TRACS – Traffic Regulation &amp; Citation System</b>   <i>React.js, Node.js, Redis, MongoDB, Nginx, PM2</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026 |
| <ul style="list-style-type: none"><li>• Engineer on Karachi’s city-wide AI traffic-enforcement platform that ingests high-volume violation data from AI-enabled CCTV cameras across the metropolis.</li><li>• Designed and implemented an on-premise GitLab CI/CD pipeline for network-restricted servers at the Central Police Office (CPO), automating builds and deployments in a secure, air-gapped environment.</li><li>• Resolved heavy read load by deploying a 3-node MongoDB replica set (1 primary, 2 read-only secondaries), improving read scalability, fault tolerance, and availability.</li><li>• Proposed and implemented Nginx as a reverse proxy/load balancer and PM2 for Node.js worker process management, increasing throughput and reliability under peak metropolitan traffic load.</li></ul> |      |
| <b>ILOS – Immutable Loan Origination System</b>   <i>React, Next.js, Node.js, PostgreSQL, Hyperledger Fabric</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025 |
| <ul style="list-style-type: none"><li>• Built an enterprise loan-origination system for UBL with fraud detection, defaulter verification, and immutable blockchain records.</li><li>• Deployed via Docker, Kubernetes, and a GitHub Actions CI/CD pipeline for seamless production rollout.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |
| <b>Blocks App – Property Tokenization Platform</b>   <i>React, Next.js, React Native Expo, Nest.js, PostgreSQL</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025 |
| <ul style="list-style-type: none"><li>• Developed a tokenization platform for HMR Builders enabling fractional property ownership and investment.</li><li>• Built the web (React/Next.js) and mobile (React Native Expo) clients with a Nest.js backend and PostgreSQL data model.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |
| <b>The Knowledge Engine</b>   <i>Python, RAG, Local LLMs, PDF Processing</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026 |
| <ul style="list-style-type: none"><li>• Built a privacy-focused, fully offline local RAG system that lets users chat with PDF documents entirely on-device.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |
| <b>AI Model Orchestrator</b>   <i>n8n, OpenRouter API, RAG, Open-Source LLMs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025 |
| <ul style="list-style-type: none"><li>• Built a dynamic AI workflow system that intelligently routes chat requests to optimal free LLMs via the OpenRouter API.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |

RESEARCH & PUBLICATIONS

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| <b>Enhancing Pneumonia Detection from Chest X-Ray Images Using ResNet-18 Deep Learning Model</b>                                                                                                                                                                                                                                                                                                                                                                                                            | 2026 |
| <ul style="list-style-type: none"><li>• Spectrum of Engineering Sciences, Vol. 4, Issue 4, pp. 863–873. Co-author: Abdul Samad. Indexed on Google Scholar.</li><li>• Proposed a ResNet-18 CNN on a 5,863-image chest X-ray dataset, achieving 98% accuracy, 98.24% precision, 97.92% recall, and 98.08% F1-score, outperforming CheXNet, VGG-19, and CNN ensembles.</li><li>• Deployed the model as a Streamlit web application for real-time diagnostics and used Grad-CAM for interpretability.</li></ul> |      |

EDUCATION

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|------------------------------------------------|--------------------------|
| <b>Sindh Madressatul Islam University</b>      | Sep 2022 – Jun 2026      |
| <i>Bachelor of Science in Computer Science</i> | <i>Karachi, Pakistan</i> |

CERTIFICATIONS & AWARDS

|                                             |   |                                                                  |
|---------------------------------------------|---|------------------------------------------------------------------|
| Best Team Member Award – Notion SMIU (2025) | • | Google AI Essentials Certificate (2024)                          |
| IT Essentials – Cisco Networking Academy    | • | Oracle Certified Associate, Java SE 8 Programmer (Expected 2026) |